

Basic Problem Statement

LWE: $B = A \cdot S + E \bmod Q$

Ring LWE: $B(x) = A(x) * S(x) + E(x) \bmod x^Q - 1$

Lattice LWE: $\vec{B} = \vec{A} \cdot \vec{S} + \vec{E}$

Encryption and Decryption in LWE

Let $(C_0, C_1) = (A, B + M)$ be the ciphertext of a message M , $\text{Enc}(M, S)$.

$$\begin{aligned}\text{Dec}(C_0, C_1, S) &= C_1 - C_0 \cdot S \\ &= (B + M) - A \cdot S \\ &= (A \cdot S + E + M) - A \cdot S \\ &= A \cdot S - A \cdot S + E + M \\ &= E + M \\ &\approx M\end{aligned}$$

Operations On Ciphertexts

Let $(C_0, C_1) = (A, B + M)$ and $(C_0', C_1') = (A', B' + M')$

Let (C_0, C_1) and (C_0', C_1')

Try component-wise addition:

$$\begin{aligned}\text{Dec}(C_0 + C_0', C_1 + C_1', S) &= (C_1 + C_1') - (C_0 + C_0') \cdot S \\ &= (B + M + B' + M') - (A + A') \cdot S \\ &= (A \cdot S + E + M + A' \cdot S + E' + M') - (A + A') \cdot S \\ &= (A \cdot S - A \cdot S) + (A' \cdot S - A' \cdot S) + E + M + E' + M' \\ &= (E + E') + (M + M')\end{aligned}$$